Quiz: Measuring Treatment Effects

Total points 8

1. Given the following statistical information of patients for a treatment arm and a control group, which one corresponds to a correct setup of a randomized control trial? **1 point**

	Treatment Arm	Control Group
Age	Mean= 55, Std = 9	Mean= 50, Std = 3
Systolic BP	Mean= 134, Std = 10.1	Mean= 132, Std = 9.2

	Treatment Arm	Control Group
Age	Mean= 60, Std =5.1	Mean= 59, Std = 5. 5
Systolic BP	Mean= 140, Std = 10.3	Mean= 139, Std = 10.1

	Treatment Arm	Control Group
Age	Mean= 30, Std = 7.1	Mean= 40, Std = 7.5
Systolic BP	Mean= 120, Std = 9.2	Mean= 140, Std = 4.9

	Treatment Arm	Control Group
Age	Mean= 61, Std = 6.7	Mean= 60, Std = 6.1
Systolic BP	Mean= 120, Std = 9.2	Mean= 140, Std = 4.9

2. You are part of a medical team trying to create an alternative treatment for patients with lung cancer. Your group performs several experiments and reports results with the following p-values. Which has the most statistically significant result? **1 point**

- p-value = 0.0001
- p-value = 0.001
- p-value = 0.5
- p-value = 0.0003

3. Given an average risk reduction (ARR) of 0.2, on average, how many people need to receive the treatment in order to benefit one of them (NNT)? **1 point**

if 100 people were treated, 20 of them would benefit from the treatment

$100 \rightarrow 20$

$x \rightarrow 1$

$x = \frac{100}{20} = \frac{10}{2} = \boxed{5}$

- [x] 5
- [ ] 10
- [ ] 20
- [ ] 0.8

4. You are studying the effect of a new treatment for heart attack, your job consists in looking at outcomes of the effect in patients, fill the unit level treatment effect column using the Neyman-Rubin causal model, and then *calculate the average treatment effect*. **1 point**

Tips:

- The event doesn't occur: 0
- The event occurs: 1
- Unit Level Treatment Effect: -1 represents a benefit, 0 represents no effect, 1 represents harm.

ID	Yi(1) Outcome Given Treatment	Yi(0) Outcome When not Given Treatment	Yi(1) - Y(0) Unit Level Treatment Effect
1	0	1	-1
2	1	0	1
3	1	1	0
4	0	0	0
5	1	0	1
6	1	1	0
7	1	0	1
8	1	0	1

$ATE = 3/8 = 0.375$

- [x] 0.375
- [ ] -0.75
- [ ] 0.75

- -0.375

5. Calculate the conditional average treatment effect applying the Two-Tree Learner method, the patient has an Age=61 and BP= 130. **1 point**

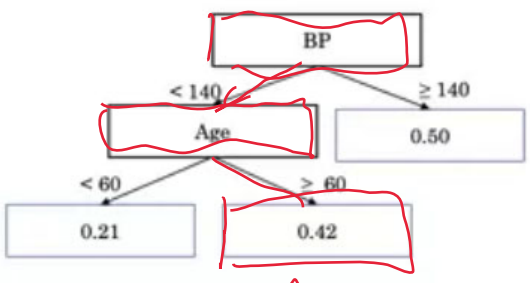
$\hat{\mu}_1(x)$ is the treatment response function.

$\hat{\mu}_0(x)$ is the control response function.

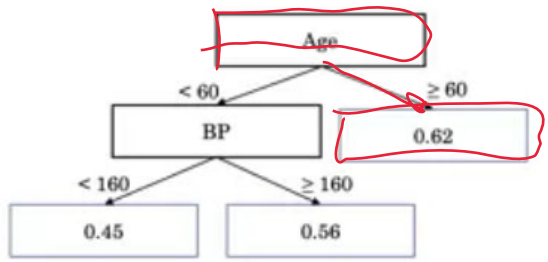
$$\mathbb{E}[Y_i(1) - Y_i(0) \mid X = x]$$

$$\mathbb{E}[Y_i \mid W = 1, X = x] - \mathbb{E}[Y_i \mid W = 0, X = x]$$

$\hat{\mu}_1(x)$



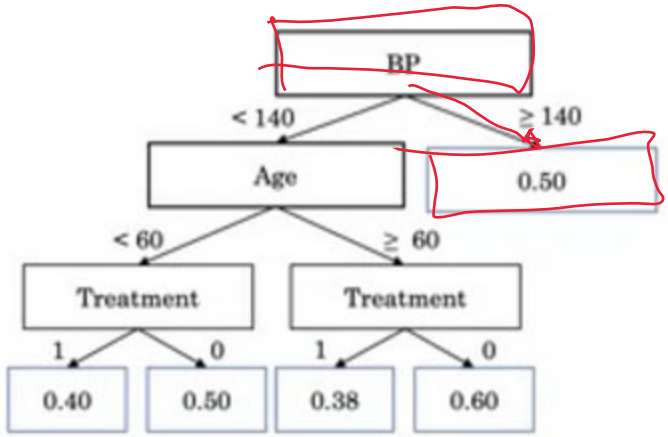
$\hat{\mu}_0(x)$



$\hat{\mu}_1(x) = 0.42$ $\hat{\mu}_0(x) = 0.62$ $CATE = \hat{\mu}_1(x) - \hat{\mu}_0(x)$
$= 0.42 - 0.62$
$= -0.2$

- 0.43
- -0.20
- 0.24
- -0.24

6. Using the S-Learner, or Single Tree, method, what is the conditional average treatment effect for a 61 year-old patient with a blood pressure (BP) of 140? **1 point**



Age=61
BP=140

- 0.10
- 0.22
- We can't estimate the conditional ATE using this S-Learner.
- 0.02

7. Which considerations are relevant to the S-Learner Method? Choose all that are correct. **1 point**

- [ ] Since the two models are using each half of the data, there are fewer samples available to learn the relationships between the features.
- [x] This model might produce a treatment effect estimate of 0 for everyone.
- [x] The Decision Tree might decide not to use the treatment feature.
- [ ] Your model is more likely to overfit your data.

8. Which considerations are relevant to the T-Learner Method? Choose all that are correct. **1 point**

- [ ] The Decision Tree might decide not to use the treatment feature.
- [ ] This model might produce a treatment effect estimate of 0 for everyone.
- [x] Since the two models are using each half of the data, there are fewer samples available to learn the relationships between the features.
- [ ] Your model is more likely to overfit your data.

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